

Drone Project - Fourth Progress Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Progress report
Primary Role	Team lead and final-integration coordinator
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Executive Summary

The fourth progress stage focused on late design comparison, three-ball carrier updates, final strategy, and future improvement planning.

The report captures the point where the team had to decide which improvements truly increased final reliability and which changes would create new risk.

Applicant Contribution Signals

- Balanced additional mechanism capability with flight and schedule risk.
- Compared late-stage design options using final-test reliability as the main criterion.
- Prepared improvement directions without compromising near-term execution.

Final Design Decisions

Late-stage drone work required restraint. Not every improvement was worth integrating if it reduced test time or increased uncertainty.

The team prioritized a final configuration that could be tested, repaired, and executed under evaluation conditions.

Future Improvement Plan

Potential improvements included cleaner payload packaging, stronger structural damping, more robust electronics layout, and more systematic test data collection.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Three-ball carrier ambition	Evaluated capability against reliability	Kept mission success central	Design judgment
Limited test time	Prioritized validated configuration	Avoided late risky changes	Risk control
Future improvements	Separated next-version ideas from final build	Protected final evaluation	Engineering planning

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.